

Introduction to Classes and Objects

The first step to understanding what an object is and how it works is to understand the relationship between a class and an object. I'll give formal definitions later, but for now, you can think of a class as a template or a blueprint that defines what an object will look like when one is created. We create objects from a class.

As an analogy, imagine if we started an on-demand cake-baking business. Being “on-demand,” we only create a cake when an order for one comes in. We specialize in Bundt cakes, and we've spent a lot of time developing the cake pan in [Figure 2-1](#) to make sure our cakes are not only tasty but also beautiful and consistent.

The pan defines what a Bundt cake will look like when we create one, but it certainly is not a cake. The pan represents our class. When an order comes in, we create a Bundt cake from our pan ([Figure 2-2](#)). The cake is an object made using the cake pan.

Using the pan, we can create any number of cakes. Our cakes could have different attributes, like different flavors, different types of frosting, and optional extras like chocolate chips, but all the cakes will come from the same cake pan.